

Pratik Shekhar

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Manipal, Karnataka

LinkedIn

GitHub

Education

Manipal Institute of Technology, Manipal

Expected May 2027

Bachelor of Technology (Electronics and Communication Engineering)

Manipal Academy of Higher Education (MAHE)

CGPA: 6.30/10

Sita Saran Memorial School

2023

CBSE Class XII — 77%

Dr. KKR Gowtham International School

2021

CBSE Class X — 91.8%

Technical Skills

Languages: Java, JavaScript, Python, C, SQL

Frontend: HTML5, CSS3, Bootstrap, React.js

Backend: Node.js, Express.js

Databases: MySQL, MongoDB

Developer Tools: Git, GitHub, Visual Studio Code

Core Concepts: Data Structures and Algorithms, Object-Oriented Programming, Full Stack Web Development

Experience

Web Developer Intern

Slash Marks IT Solutions

Jun 2026 – Jul 2026

- Developed responsive websites using HTML, CSS, JavaScript, React.js and Node.js.
- Collaborated with the web development team to deliver assigned web projects.
- Improved UI layouts and optimized webpage assets for better user experience.

Projects

Smart Traffic Light System with Emergency Priority

STM32F401, Embedded C, RFID

- Designed an STM32F401-based smart traffic controller with dynamic signal timing.
- Integrated RFID-based emergency vehicle detection for automatic signal preemption.
- Implemented interrupt-driven control ensuring low-latency emergency response.

Cross-Layer Application-Aware Routing & Traffic Policing

MATLAB, Computer Networks

- Developed a MATLAB-based cross-layer routing simulator integrating Network, Transport and Application layers.
- Implemented Dijkstra's shortest path algorithm with dynamic topology updates.
- Designed Leaky Bucket traffic policing for congestion detection and application-aware routing.

Analog Audio Equalizer

LTspice, Analog Electronics

- Designed a multi-band analog audio equalizer using operational amplifiers and active RC filter networks.
- Analyzed frequency response and stability using LTspice simulations.

Areas of Interest

Software Development • Data Structures & Algorithms • Web Development • AI Automation & Prompt Engineering

Languages

English Hindi